University of BRISTOL

Centre for Academic Language and Development

Foundations of Computer Programming - LANG00052

Week 03

Flowchart

Syed Murad

Learning outcomes

- Understand the fundamental components of a flowchart
- Identify Flowchart Symbols
- Learn how to interpret and follow the logic of a flowchart
- Learn how to create a flowchart for a simple process
- Practice designing and creating flowcharts

Flowchart

A graphical representation of the sequence of operations in an information system or program.

- Program flowcharts show the sequence of instructions in a single program or subroutine.
- Flowcharts employ various symbols to represent different components of the program or subroutine.

Flowchart Symbols



Denotes the start and end of an algorithm

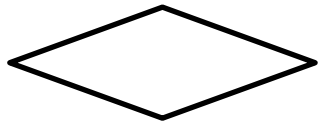


Denotes a process to be carried out



Denotes a sub-process

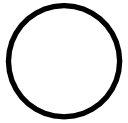
Flowchart Symbols



Denotes a decision to be made

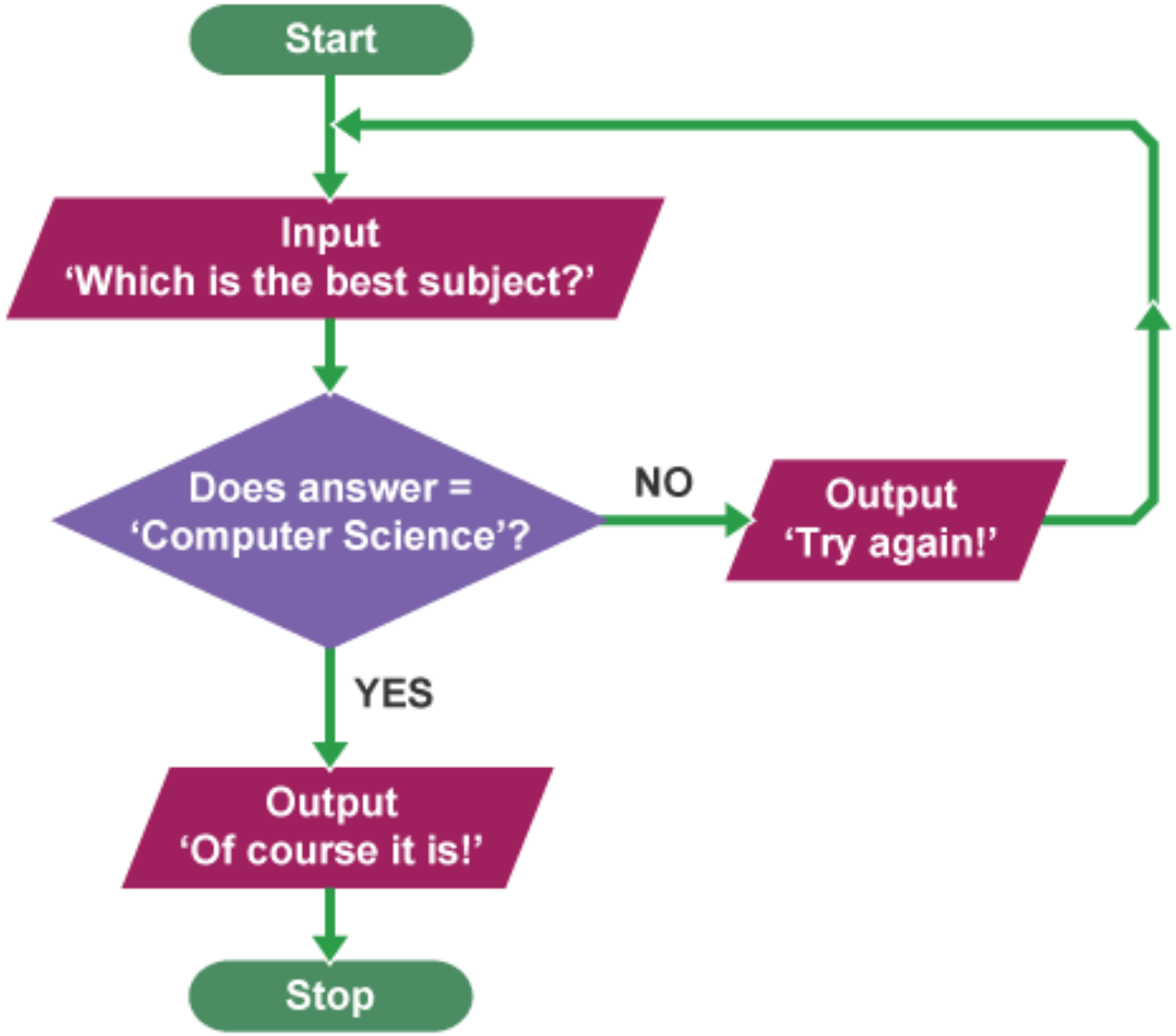


Denotes input or output

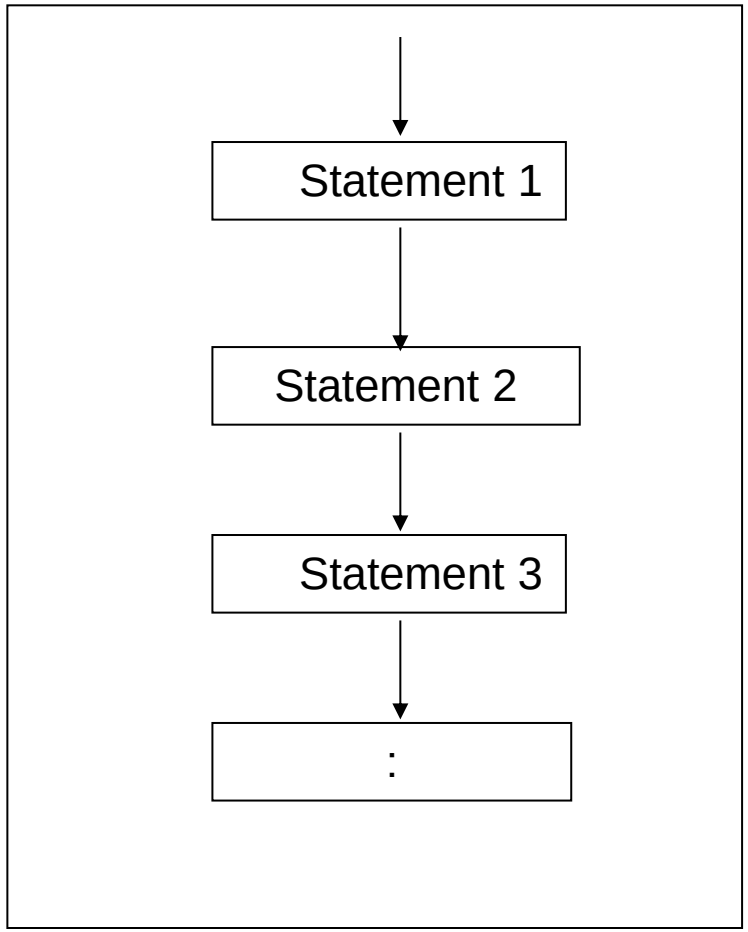


Denotes a connection to part of a flowchart that cannot easily be linked using an unbroken flow arrow

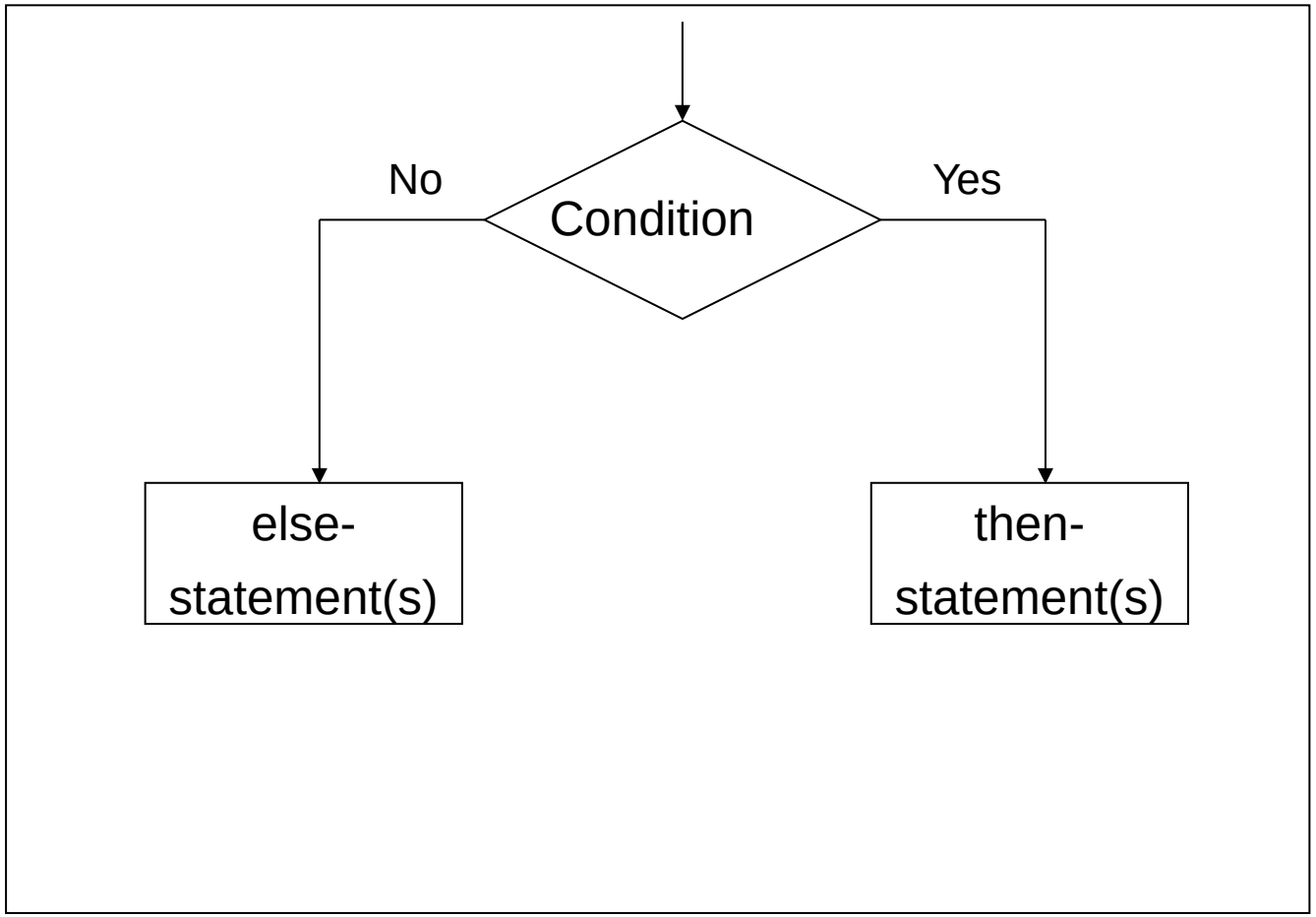
Shows the logical flow of the program


Start
Input
'Which is the best subject?'
Does answer =
'Computer Science'?
NO
Output
'Try again!'
YES
Output
'Of course it is!'
Stop

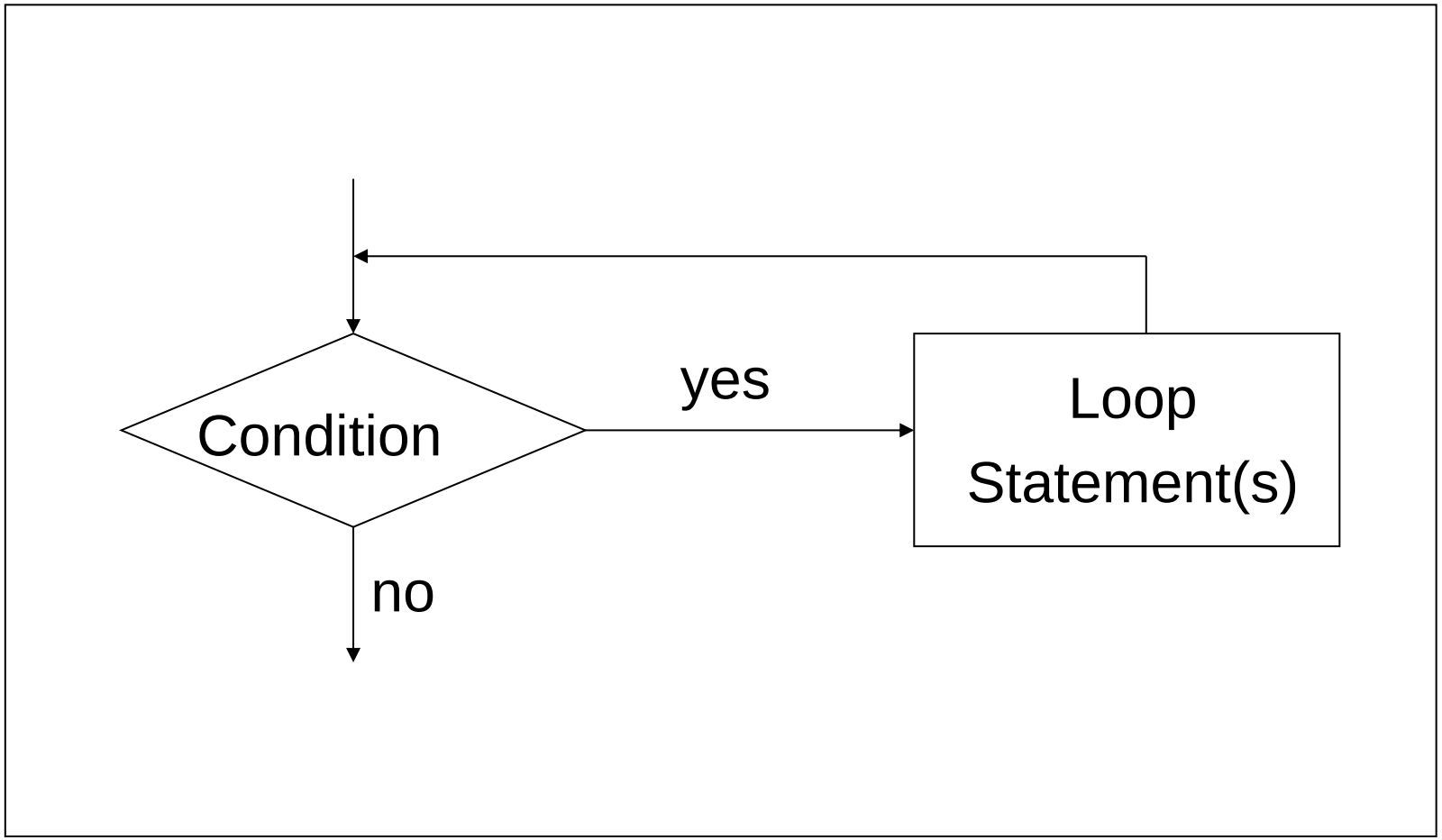
Flowchart – sequence control structure



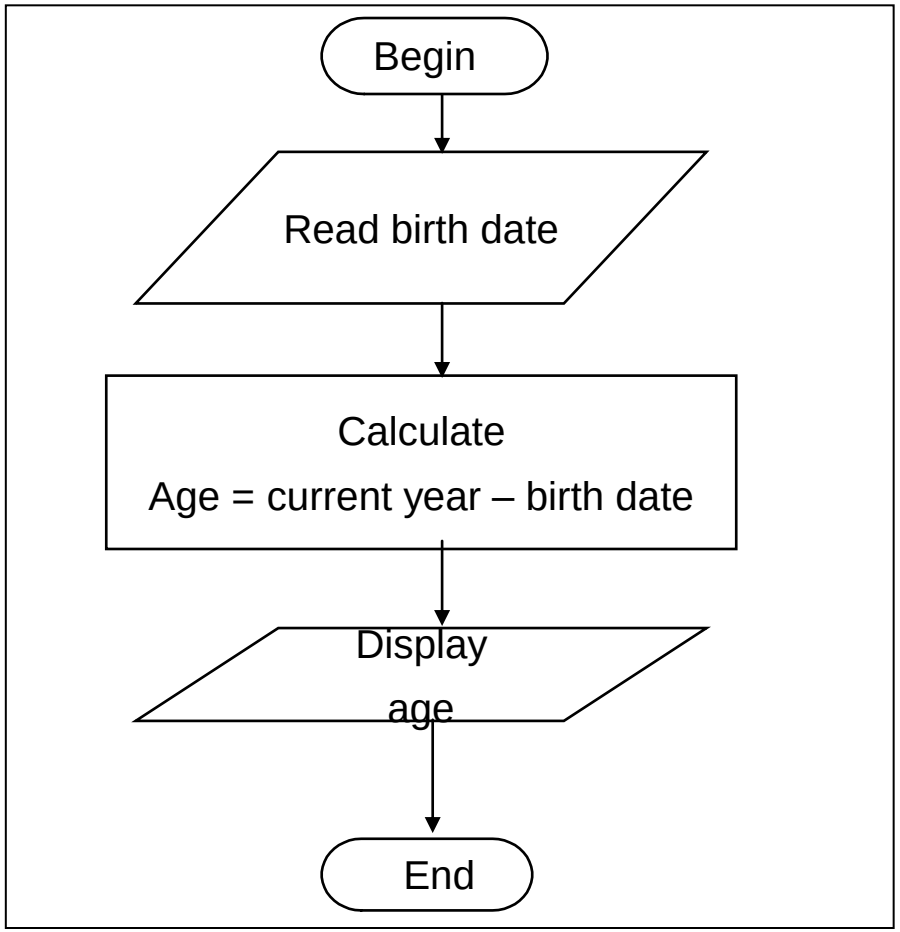
Flowchart – selection control structure



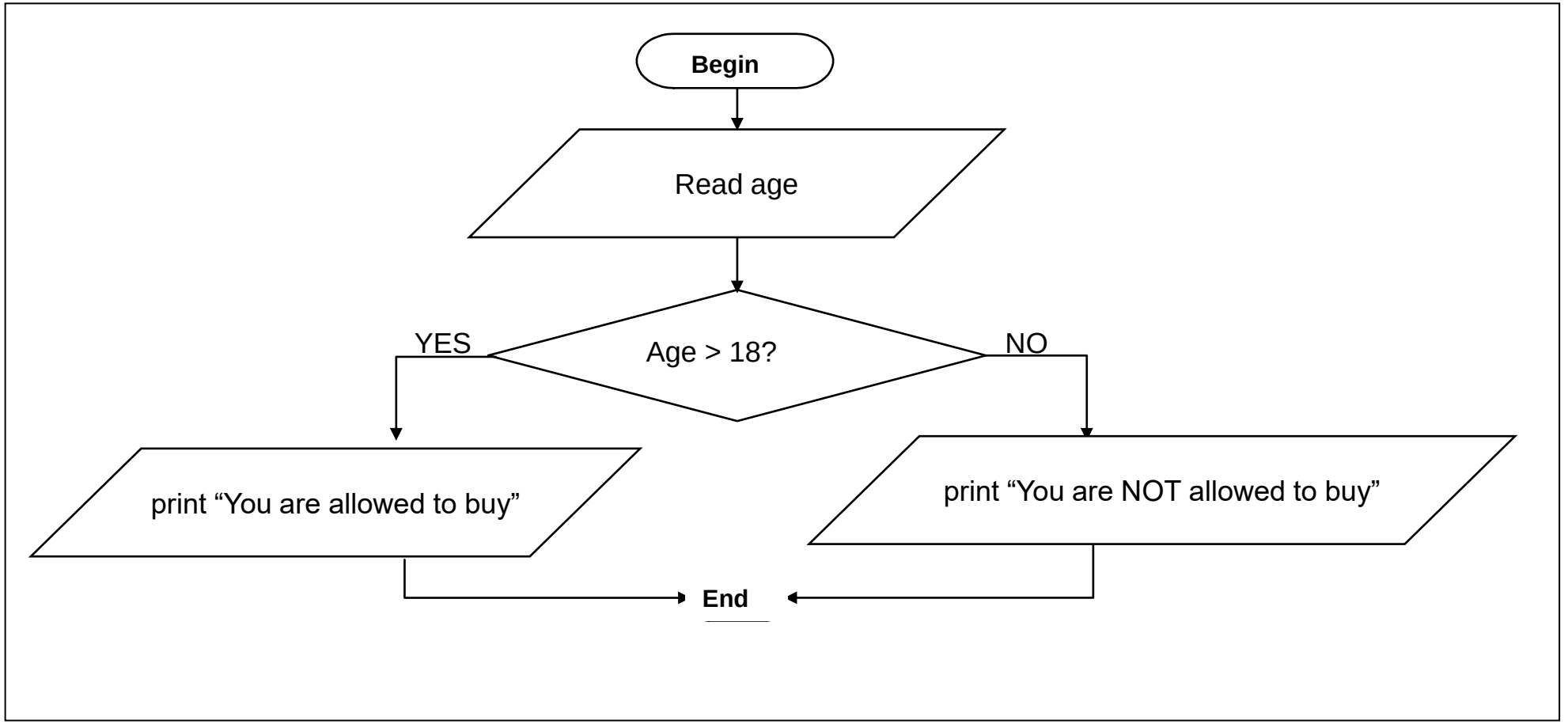
Flowchart – repetition control structure



Example 1


Begin
Read birth date
Calculate
Age = current year – birth date
Display
age
End

Example 2


Begin
Read age
Age > 18?
YES
NO
print “You are allowed to buy”
print “You are NOT allowed to buy”
End

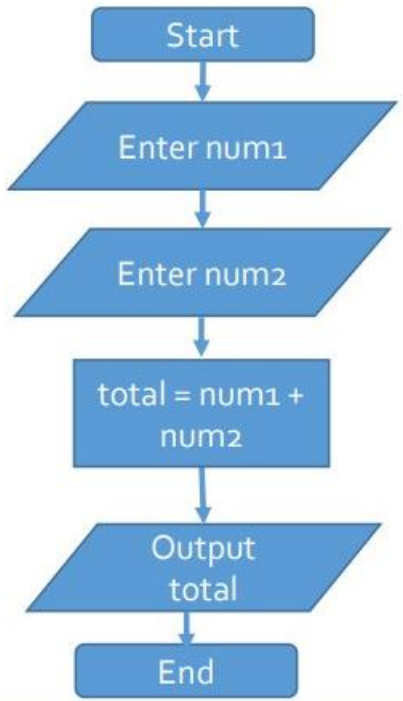
Design a flowchart that prompts the user to input two numbers, performs an addition operation, and then displays the resulting sum.

Now write pseudocode!

Written Description

Enter first number
Enter second number
Add together to get the total
Output the total

Flowchart



Pseudocode

```
SEND 'Enter first number' TO DISPLAY
RECEIVE num1 FROM (INTEGER) KEYBOARD
SEND 'Enter second number' TO DISPLAY
RECEIVE num2 FROM (INTEGER) KEYBOARD

SET total TO num1 + num2
SEND total TO DISPLAY

END
```

Why do we create flowcharts, and what are their benefits?

Time for questions